

COMPUTING SYSTEM PROJECT DIGITAL LOCK EXECUTIVE SUMMARY

Carlos Eduardo Baniski, Gustavo Guse Martins, Kevin Kurpias Rodrigues, Vitor Rafael Linhares
de Macedo

Ponta Grossa April, 2025

TABLE OF CONTENTS

1. INTRODUCTION
2. JUSTIFICATIONS
3. OBJECTIVES 3.1 General Objective 3.2 Specific Objectives
4. ELECTRICAL DESIGN
5. PHYSICAL DESIGN
6. COMMUNICATION DESIGN
7. SOFTWARE MODELING
8. DATA MODELING
9. OPERATION MANUALS
10. ELECTRICAL DIAGRAM

1. INTRODUCTION

This document details the proposal for developing a smart and autonomous electronic lock prototype. The project consists of creating a high-security access control system managed by an ESP32 microcontroller, integrating three distinct and complementary local authentication methods. The physical locking of the door will be performed by a 12V Motor acting directly on the latch mechanism. To ensure system reliability, the project incorporates an uninterruptible power supply system based on an 18650 UPS Module, alongside an LM2596 Voltage Regulator.

2. JUSTIFICATIONS

The transition from purely mechanical locking systems to electronic access control solutions has become an unavoidable requirement in modern physical security and automation projects. In this context, the development of this electronic lock based on local authentication is technically and operationally justified by four fundamental pillars of embedded engineering. Firstly, the project stands out for the convergence of convenience and high operational availability through biometric and digital redundancy. The integration of two completely independent input methods—digital biometrics and numerical password—ensures the user does not suffer accidental lockouts. In case of a forgotten numerical PIN, biometric reading via fingerprint acts as an immediate, secure, and non-transferable contingency system. Secondly, the proposed architecture solves a critical flaw common to entry-level commercial products: energy fragility based on conventional alkaline batteries. By implementing a dedicated UPS module powered by high-capacity lithium cells (18650 standard), the prototype gains autonomous resilience. Operating as an internal DC UPS, the system continuously preserves security, keeping access control active even during instabilities or prolonged outages in the electrical supply network. Furthermore, selecting the ESP32 microcontroller as the logical unit goes beyond processing local validations in real-time ; it embeds immense expansion potential for the Internet of Things (IoT) ecosystem. With native support for Wi-Fi networks and Bluetooth connectivity, the hardware is natively prepared for scalability, enabling future updates such as remote management via mobile apps, cloud access auditing, and direct communication with open-source home automation hubs (like Home Assistant). Finally, from the perspective of electrical circuit protection and stability, including a switching voltage regulator (LM2596 step-down) is imperative. This component isolates and stabilizes the power supply, ensuring that reading modules operating at low voltages (3.3V/5V) are completely shielded against transient inductive noise and fluctuations generated by the current peak when triggering the 12V electromechanical motor.

3. OBJECTIVES

3.1 General Objective

Develop, integrate, and validate a functional, robust, and energy-autonomous prototype of a

multi-local-access electronic lock capable of managing the mechanical unlocking release securely through validation via digital biometrics or a matrix numerical password.

3.2 Specific Objectives

- **Hardware Module Integration:** Size and electrically connect the ESP32 microcontroller to the JP300 biometric reader and the matrix keypad, correctly mapping the GPIO pin interfaces and ensuring signal level compatibility.
- **Firmware Development:** Program the low-level logic in the ESP32 to register, store in non-volatile memory, and validate access credentials in microseconds.
- **Electromechanical Power Control:** Design and implement the switching and isolation circuit to safely drive the 12V DC Motor from the low-power logic signal emitted by the microcontroller.
- **Energy Management:** Structure the continuous power supply circuit with the 18650 UPS module and calibrate the LM2596 switching regulator to provide rectified and stable voltages for the logic part and sensors.

4. ELECTRICAL DESIGN

The electrical design of the system was developed following an approach of dividing power domains, essential to mitigate inductive noise and thermal fluctuations caused by mechanical actuators on sensitive integrated circuits. The electrical architecture is subdivided into three main buses:

1. **Charge and Backup Bus (UPS):** Centered on the 18650 UPS Module, which manages the charge and float cycles of the lithium-ion batteries. This block receives external power and provides a stable voltage line for the system. In case of a main power grid failure, the batteries take over the load with zero transfer time.
2. **Power/Force Bus (12V):** A direct line that powers the DC motor responsible for physically locking the door. The activation of this bus is isolated by means of an optocoupled relay or N-channel Field Effect Transistor (MOSFET), configured as a static switch. This prevents the inductive kickback from the motor coil from propagating to the digital line.
3. **Regulated Logic Bus (5V and 3.3V):** The main circuit voltage is stepped down with high efficiency through the LM2596 Buck step-down converter. The regulated 5V output directly powers the JP300 biometric reader and the ESP32 power pins. The ESP32's internal regulator derives the 3.3V line needed for the operation of its GPIO pins and internal buses.

5. PHYSICAL DESIGN

The physical design and 3D modeling of the prototype focus on the safe encapsulation of electronic components and the mechanical coupling of the actuator to the lock's latch. The external structure was designed to be manufactured in high-density acrylic or 3D printing (ABS polymer), ensuring mechanical protection and thermal and electrical insulation.

The front panel ergonomically exposes the human-machine interaction interfaces: the JP300 biometric reader's optical sensor is positioned at the top to facilitate fingerprint reading, while the matrix keypad is arranged just below for comfortable PIN typing.

Internally, the box has dedicated supports for fixing the ESP32 printed circuit board (PCB), the LM2596 regulator module, and the 18650 UPS module, minimizing electrical conductor length and preventing short circuits due to vibration. The 12V Motor is rigidly aligned with the lock's mechanical core, where a rotating shaft or cam transforms the motor's kinematic movement into the physical retraction of the locking latch.

6. COMMUNICATION DESIGN

To ensure integrity and synchronism in information exchange between the central microcontroller and peripherals, the system adopts a mixed communication architecture, combining asynchronous serial protocols, pulse-width modulation, and discrete signal scanning.

The interface between the ESP32 and the AS608 biometric sensor is governed by the physical UART (Universal Asynchronous Receiver-Transmitter) protocol. The firmware implementation instantiates a dedicated hardware serial bus (UART Channel 2, via `HardwareSerial`) on pins GPIO 25 (RX) and GPIO 26 (TX).

To ensure immunity to electromagnetic noise from motor switching, the communication operates at a set baud rate of 57,600 bps (`SERIAL_8N1` standard). Heavy vector processing occurs decentralized in the AS608's internal DSP chip, offloading the ESP32 core, which only requests quick searches via serial commands (`fingerFastSearch()`).

Communication with the 4x4 membrane keypad bypasses dedicated ICs, using the Active Matrix Scanning technique controlled at runtime by the `Keypad.h` library. The bus consumes 8 contiguous digital paths of the microcontroller:

- **Rows (Active outputs):** Mapped to GPIOs 19, 18, 5, and 17;
- **Columns (Inputs with internal Pull-up):** Mapped to GPIOs 2, 0, 4, and 16.

Unlike logic interfaces, power and monitoring peripherals operate via modulation and physical contact dry switches:

- **Modulated Power Control:** The 12V motor drive is not binary and symmetrical. The system employs a Pulse Width Modulation (PWM) technique through the `analogWrite()` instruction applied to GPIO 32. This technique saturates the MOSFET Gate with a variable Duty Cycle, allowing calibration of the mechanical torque and angular speed of the lock (currently empirically set at a factor of 145 out of 255) to prevent gear damage.
- **Monitoring Sensors:** The limit microswitch (GPIO 14) and the Reed Switch magnetic sensor (GPIO 22) operate with the technique of internal pull-up resistors activated via firmware (`INPUT_PULLUP`). They ground the microcontroller pins sending a low logic level (LOW) only when the physical state of "locked" or "door closed" is reached.

7. SOFTWARE MODELING

7.1 Use Case Diagram The diagram maps available actions for the common User

(authentication for opening) and for the Administrator (system maintenance and management).

7.2 Class Diagram The object-oriented structure maps physical hardware modules as abstracted classes within the firmware code.

7.3 Flowchart (Activity Diagram) The flow represents the continuous execution cycle (main loop) executed by the microcontroller.

8. DATA MODELING

Access credential management was architected to resist direct memory extractions from the chip and ensure resilience against critical power failures. Data persistence occurs in the ESP32's native Flash memory partition, using the NVS (Non-Volatile Storage) subsystem, abstracted by the Preferences class. The logical structure adopts the Key-Value Store data dictionary model.

Key	Data Type	Functional Description
senha_mestre	String	Cryptographic signature (SHA-256 Hash) corresponding to the Administrator Password (implemented factory default value).
user_0 to user_4	String	Logic array (slots) destined for storing the password Hashes of common users.
proximo_slot	Integer (int)	Pointer variable indicating the index of the next free user slot for registration under the modular Round-Robin algorithm (% TOTAL_USER_SLOTS).

As a primary defense mechanism, the code does not store matrices or literal alphanumeric strings on the disk. The saving flow requires passing through the library based on the Mbed TLS protocol specification (mbedtls/md.h).

Any typed character string is converted in volatile memory into an arbitrary-sized minutiae vector, injected into the SHA-256 structural algorithm, processed in the CPU's cryptographic context, and finalized as a 64-character hexadecimal string. Access validation requires the live

computational calculation of the typed hash block and the logical equivalence comparison against the stored NVS database.

Furthermore, the firmware preventatively disables the RTC_CNTL_BROWN_OUT_REG register via pointer in machine memory (addressed by the soc/soc.h library). This prevents the processor's abrupt restart in case of transient drops in the electrical supply voltage derived from the high consumption during the initial triggering of the inductive motor.

9. OPERATION MANUALS

To guarantee correct human-computer interaction (HCI) and the prototype's safe operation, a set of operational procedures was established. The system uses a visual feedback interface (Status LED) to guide the user during machine state transitions.

Initial system configuration parameters are defined directly in the firmware:

- **Default Master Password:** 123456
- **Confirmation Key:** # (Submits the data buffer for processing).
- **Cancel/Reset Key:** * (Clears the volatile typing memory and aborts ongoing operations).

The release of the electromechanical lock can be performed through two authentication branches:

1. **Biometric Access:** The user places the previously registered finger on the AS608 optical reader. Processing occurs in real-time (less than 1 second). In case of positive validation (FINGERPRINT_OK), the status LED is triggered and the motor is unlocked.
2. **Matrix Keypad Access:** The user inputs their numerical password (or Master Password) and presses the # key. If the generated hash matches a valid credential in the NVS database, the door unlocks. In case of authentication failure, the system emits a visual alert (LED blinks 4 times at high frequency).

To mitigate human errors related to forgetting, the system implements an auto-lock routine based on timing and physical sensing:

1. After the opening event, the microcontroller starts a 10-second delay timer.
2. Once the time is up, the firmware reads the magnetic door sensor (MAG_PIN).
3. If the door is detected as closed (LOW), the motor is triggered to the locking position.
4. If the door is detected as open (HIGH), the system enters an alert state (intermittently blinking LED) and suspends the main loop execution until physical contact with the frame is reestablished, ensuring the latch is not pushed out while the door is open.

Access credential management is restricted to the system administrator and requires prior Master Password authentication.

New User Registration (Numerical Password):

1. The administrator inputs the command code BCDA and presses #.
2. Upon the solid lighting of the status LED, the Master Password is input, followed by #.
3. With positive validation (2 LED blinks), the administrator types the desired new password and presses #.
4. The system stores the new hash in the next free memory space (holding up to 5 common users) and confirms the operation with 3 long LED blinks.

New Fingerprint Registration:

1. The administrator inputs the command code ABCD and presses #.
2. After authorizing the operation with the Master Password (confirmed by 2 LED blinks), the system enters biometric listening mode.
3. The user places the finger on the reader for the first minutiae capture.
4. After the initial read, the finger must be removed for 2 seconds and repositioned exactly the same way for cross-validation capture.
5. If the biometric matrices are compatible, the system saves the model to the AS608 chip and emits 3 status LED blinks.

Biometric Database Cleanup (Reset):

1. The administrator inputs the emergency code 9999 and presses #.
2. The microcontroller sends the global deletion command (`emptyDatabase()`) via the serial bus to the AS608 module.
3. The system confirms the biometric memory formatting with 3 slow status LED blinks.

10. ELECTRICAL DIAGRAM

To assist with assembly and maintenance, the electrical diagram is provided below. Pay attention to use the names of the red connections as a reference, rather than the ones written inside the chips, due to differences in nomenclature.